Kacem Mathlouthi

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Education

National Institute of Applied Science and Technology (INSAT) *Sept 2022 – May 2027*
Engineering Degree in Software Engineering

Experience

Software Engineer *On Site*
Callab AI (YC P26) – Conversational Voice AI Agents *Jun 2025 – Present*

- Built **Web Call** feature end-to-end across full stack, delivering browser-based voice calls that eliminated telephony providers.
- Developed and published **webcall-widget** to **npm**, enabling seamless third-party voice agent integration.
- Engineered high-performance Call Recorder microservice in **Rust**, handling concurrent recording sessions.
- Reduced average latency by **250ms** through audio mixing and noise cancellation optimization, significantly improving UX.
- Built **LLM-powered** post-call analysis pipeline to automatically extract actionable insights and conversation outcomes.

Google Summer of Code Developer *Remote*
NRNB (National Resource for Network Biology) *May 2025 – Sep 2025*

- Collaborated with the VCell team to develop and deploy **VCell-AI**, an AI agent platform with **function-calling** capabilities, featuring a **Next.js** frontend and **FastAPI** backend, enabling researchers to query, explore, and create biological models.
- Integrated **RAG** system using **Qdrant** to provide contextual information about VCell software, tutorials, and guidelines.
- Enabled AI agent to analyze diagrams and VCML files using multimodal capabilities, supporting biological model interpretation.
- Implemented **LLM observability** via **Langfuse** to track agent interactions, monitor performance, and improve reliability.

Machine Learning Engineering Intern *Tunis, Tunisia*
Orange *Aug 2024 – Oct 2024*

- Developed lightweight **CNN model** for plant disease classification optimized for **edge devices** using **knowledge distillation**.
- Built **AI agentic workflow** for automated report generation, streamlining data collection, analysis, and research extraction.
- Achieved **4th place** in Orange Summer Challenge, pitching the solution to both technical and non-technical stakeholders.

Publications

Generalization Measures under Controlled Covariate Shift: A Regime-Aware Benchmark
Accepted at Transactions on Machine Learning Research (TMLR) *OpenReview*

- Evaluated 40+ generalization measures on CIFAR-10-C/P under controlled covariate shift, finding that predictivity is strongly regime-dependent and that IID performance does not reliably transfer to distribution shift or model selection.

Projects

Metis AI: *FastAPI, React, Celery, Redis, PostgreSQL, GitHub App, GCP* *Live Demo*

- Built and deployed a full-stack **GitHub App** that automates pull request reviews using **AI agents** in a sandboxed environment.
- Designed asynchronous processing with **Celery + Redis** to handle webhook-driven review workflows and background issue-to-PR coding tasks.

Achievements and Recognitions

1st place at **Orange x Hexabot Conversational AI Hackathon** *devpost/x2x-Submission*

- Built extensions for Hexabot visual editor, enabling multimodal capabilities like text-to-speech, speech-to-text, and vision.

2nd place at **IEEE TSYP 12 CS Technical Challenge** *Smartshield-INSAT*

- Earned 2nd place at the Tunisian Students and Young Professionals Computer Science Challenge by creating an AI-powered platform for automated cybersecurity threat detection and response, Smartshield.

2nd place at **Artificial Intelligence National Summit (AINS 2.0)** *AINS-ML.Guide*

- Implemented a Multi-LLM Agent System to help engineers define, evaluate, and solve machine learning problems.

Technologies

Languages: Python, Typescript, Go, Rust, SQL

Tools & Technologies: Git, Linux, Docker, NGINX, Azure, GitHub/GitLab, MongoDB, PostgreSQL, Vector Databases, React, Next.js, FastAPI, Flask, Express.js, LangChain, RAG Systems, LLM Workflows, Langfuse